

RAMNICU VALCEA, Romania

WMO: 153460

Lat: 45.0889N

Lon: 24.3628E

Elev: 238

StdP: 98.49

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.4	-14.2	1.1	-7.8	-11.7	1.4	-5.8	6.2	3.5	4.9	2.6	0.8	350	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	33.4	21.0	31.6	20.6	30.0	20.2	22.6	30.2	21.7	29.1	20.9	28.0	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.3	25.6	19.2	14.4	24.8	18.5	13.7	24.0	67.5	30.2	64.2	29.2	61.4	28.0	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.3	3.7	-15.2	35.4	3.2	2.1	-17.6	37.0	-19.5	38.2	-21.3	39.4	-23.7	41.0		
(5)			-15.4	24.3	3.2	1.1	-17.8	25.0	-19.7	25.7	-21.5	26.3	-23.8	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-0.1	2.3	6.7	12.3	17.2	20.8	22.8	22.7	17.7	12.1	6.6	1.3		
	DBStd	8.90	4.39	4.70	4.25	3.65	3.25	3.04	2.77	2.97	3.61	3.98	4.16	4.14		
	HDD10.0	1077	312	218	117	18	1	0	0	0	25	118	269			
	HDD18.3	2771	570	450	361	182	61	12	1	3	54	196	353	527		
	CDD10.0	1775	0	1	15	88	223	323	396	394	231	89	15	0		
	CDD18.3	427	0	0	0	2	24	85	139	139	36	2	0	0		
	CDH23.3	3962	0	0	1	37	245	773	1307	1250	321	28	0	0		
	CDH26.7	1437	0	0	0	4	42	253	523	521	91	2	0	0		
(14) Wind	WSAvg	1.6	1.3	1.5	1.8	1.9	1.8	1.8	1.8	1.7	1.6	1.5	1.4	1.3		
(15) Precipitation	PrecAvg	732	39	32	39	60	84	90	87	80	64	57	49	49		
	PrecMax	1185	91	74	88	160	190	159	225	217	147	142	136	122		
	PrecMin	401	4	3	2	4	31	9	22	10	4	1	2	0		
	PrecStd	175	23	18	24	31	41	40	49	48	40	36	31	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	17.2	21.8	27.3	29.7	33.7	36.4	35.8	32.1	26.3	20.2	13.2		
		MCWB	7.6	9.9	12.0	15.6	18.0	21.9	20.6	20.9	19.3	16.7	13.2	8.6		
	2%	DB	10.1	13.8	19.0	23.8	27.8	31.4	33.4	33.4	29.4	23.5	17.1	10.4		
		MCWB	6.4	8.2	10.5	14.4	17.6	20.9	21.1	20.9	18.4	15.7	12.2	7.5		
	5%	DB	7.6	11.2	16.5	21.3	26.2	29.5	31.4	31.6	26.9	21.0	14.6	8.4		
		MCWB	5.1	6.7	9.4	12.9	17.1	20.3	20.9	20.6	17.6	14.7	10.7	6.3		
	10%	DB	5.6	8.8	14.1	19.3	24.3	27.8	29.7	29.7	24.8	18.6	12.6	6.8		
		MCWB	3.7	5.6	8.2	12.0	16.4	19.5	20.4	20.1	16.9	13.5	10.0	5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	10.5	12.6	16.5	20.0	23.1	23.4	24.1	20.6	18.3	14.2	9.6		
		MCDB	11.8	16.1	19.7	24.6	27.0	31.1	31.2	31.0	28.8	24.1	18.3	11.8		
	2%	WB	7.0	8.7	11.1	15.0	18.7	21.9	22.5	22.6	19.2	16.5	12.8	8.1		
		MCDB	9.4	13.0	17.3	22.5	25.6	29.4	30.0	30.3	26.8	21.4	15.9	9.8		
	5%	WB	5.4	7.3	10.0	13.7	17.8	21.0	21.7	21.5	18.3	15.3	11.5	6.6		
		MCDB	7.2	10.2	15.6	19.8	24.5	28.0	29.1	29.0	24.9	19.8	13.9	8.1		
	10%	WB	3.9	5.9	8.8	12.7	16.9	20.1	20.9	20.6	17.5	14.1	10.3	5.4		
		MCDB	5.4	8.4	13.3	17.8	22.7	26.5	28.0	27.8	23.1	17.9	12.4	6.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.5	10.1	10.7	11.3	11.5	12.0	12.0	11.1	10.1	8.0	7.0		
		MCDBR	11.0	12.8	14.7	14.5	14.1	14.2	14.6	14.7	14.6	13.8	11.8	9.9		
	5% WB	MCWBR	7.2	7.7	7.7	6.6	6.0	6.0	5.5	5.2	5.7	6.9	6.9	6.7		
		MCWBR	9.9	11.7	13.1	12.9	12.8	13.0	13.0	13.0	12.8	12.4	10.2	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.359	0.400	0.463	0.442	0.443	0.458	0.477	0.428	0.387	0.360	0.327		
	taud		2.310	2.227	2.183	2.035	2.166	2.205	2.172	2.109	2.210	2.314	2.333	2.341		
	Ebn at Noon		770	806	822	797	831	830	811	777	784	766	720	725		
	Edh at Noon		81	106	127	160	145	140	144	147	123	97	80	72		
(44) All-Sky Solar Radiation	RadAvg		1.40	2.23	3.37	4.47	5.46	5.98	6.04	5.40	3.93	2.65	1.52	1.15		
	RadStd		0.17	0.31	0.38	0.47	0.48	0.49	0.46	0.43	0.36	0.34	0.18	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.64	N/A	N/A	N/A	N/A	N/A	-114	-180	+136	+71
(47) Regional (78 neighbors)		+0.57	N/A	N/A	+0.80	N/A	N/A	N/A	-161	+82	+39

Nomenclature: See separate page